Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-11 Emerged Berm 4% - Test 2.experimentvelocity.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.344 Hz*
Freq. upper bound:	*6.875 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*23*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*28*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *33.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1251*	*0.04215*
Peak wave period T_P (s):	*1.455*	*1.455*
Energy wave period $T_{-1,0}$ (s):	*1.343*	*1.214*
Mean wave period $T_{0,1}$ (s):	*1.26*	*0.9809*
Mean wave period $T_{0,2}$ (s):	*1.201*	*0.8609*
Spectral width (Broadness):	*0.7125*	*0.8073*
Spectral width (Narrowness):	*0.3176*	*0.5463*

Time Domain Analysis

Overall amplitude reflection coefficient: *33.7%*

	Incident	**Reflected**
Number of waves:	931	1136
Mean wave height H_m (m):	*0.08072*	*0.02749*
Mean wave period T_m (s):	*1.274*	*1.045*
Sig. wave height H_s (m):	*0.1217*	*0.04158*
$T_{H_{1/3}}$ (s):	*1.389*	*1.262*
H_{max} (m):	*0.1736*	*0.07388*
T_{Hmax} (s):	*1.569*	*1.451*
$H_{1/10}$ (m):	*0.1441*	*0.05098*
$H_{1/50}$ (m):	*0.1636*	*0.06042*
$H_{1/100}$ (m)	*0.1694*	*0.06387*
$H_{1/250}$ (m):	*0.1724*	*0.068*
$H_{0.1\%}$ (m):	*Not enough data*	*0.07294*
$H_{1\%}$ (m):	*0.1644*	*0.05885*
$H_{2\%}$ (m):	*0.1541*	*0.05562*
$H_{10\%}$ (m)	*0.1287*	*0.04448*
Groupiness factor GF:	*0.9055*	*0.9986*
Skewness b1:	*0.3789*	*0.3224*
Kurtosis b2:	*2.791*	*3.01*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-11 Emerged Berm 4% - Test 2.experimentvelocity.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*24*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *34.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1383*	*0.04796*
Peak wave period T_P (s):	*1.524*	*1.6*
Energy wave period $T_{-1,0}$ (s):	*1.403*	*1.251*
Mean wave period $T_{0,1}$ (s):	*1.301*	*0.9822*
Mean wave period $T_{0,2}$ (s):	*1.228*	*0.8522*
Spectral width (Broadness):	*0.749*	*0.8115*
Spectral width (Narrowness):	*0.3489*	*0.573*

Time Domain Analysis

Overall amplitude reflection coefficient: *34.7%*

	Incident	**Reflected**
Number of waves:	943	1144
Mean wave height H_m (m):	*0.08994*	*0.03206*
Mean wave period T_m (s):	*1.323*	*1.09*
Sig. wave height H_s (m):	*0.1354*	*0.04824*
$T_{H_{1/3}}$ (s):	*1.462*	*1.316*
H_{max} (m):	*0.1976*	*0.07611*
T_{Hmax} (s):	*1.597*	*1.353*
$H_{1/10}$ (m):	*0.1593*	*0.05826*
$H_{1/50}$ (m):	*0.1814*	*0.0691*
$H_{1/100}$ (m)	*0.1871*	*0.07183*
$H_{1/250}$ (m):	*0.1937*	*0.07441*
$H_{0.1\%}$ (m):	*Not enough data*	*0.07611*
$H_{1\%}$ (m):	*0.1807*	*0.06896*
$H_{2\%}$ (m):	*0.1724*	*0.06357*
$H_{10\%}$ (m)	*0.143*	*0.05139*
Groupiness factor GF:	*0.8861*	*0.9981*
Skewness b1:	*0.4649*	*0.2684*
Kurtosis b2:	*2.755*	*2.97*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-11 Emerged Berm 4% - Test 2.experiment.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.320 Hz*
Freq. upper bound:	*6.406 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*30*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *34.9%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1449*	*0.05056*
Peak wave period T_P (s):	*1.561*	*1.641*
Energy wave period $T_{-1,0}$ (s):	*1.453*	*1.318*
Mean wave period $T_{0,1}$ (s):	*1.339*	*1.008*
Mean wave period $T_{0,2}$ (s):	*1.258*	*0.864*
Spectral width (Broadness):	*0.766*	*0.8196*
Spectral width (Narrowness):	*0.3649*	*0.6013*

Time Domain Analysis

Overall amplitude reflection coefficient: *34.8%*

	Incident	Reflected
Number of waves:	948	1193
Mean wave height H_m (m):	*0.09626*	*0.03364*
Mean wave period T_m (s):	*1.384*	*1.1*
Sig. wave height H_s (m):	*0.143*	*0.05059*
$T_{H_{1/3}}$ (s):	*1.533*	*1.364*
H_{max} (m):	*0.2128*	*0.08423*
T_{Hmax} (s):	*1.543*	*1.777*
$H_{1/10}$ (m):	*0.1663*	*0.0626*
$H_{1/50}$ (m):	*0.1906*	*0.075*
$H_{1/100}$ (m)	*0.1983*	*0.07866*
$H_{1/250}$ (m):	*0.2057*	*0.08113*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08391*
$H_{1\%}$ (m):	*0.1894*	*0.07518*
$H_{2\%}$ (m):	*0.1772*	*0.06797*
$H_{10\%}$ (m)	*0.1502*	*0.05421*
Groupiness factor GF:	*0.8725*	*1.011*
Skewness b1:	*0.5051*	*0.2752*
Kurtosis b2:	*2.721*	*3.075*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-11 Emerged Berm 4% - Test 2.experimentvelocity.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.313 Hz*
Freq. upper bound:	*6.250 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*31*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *34.2%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1573*	*0.05372*
Peak wave period T_P (s):	*1.6*	*1.73*
Energy wave period $T_{-1,0}$ (s):	*1.509*	*1.373*
Mean wave period $T_{0,1}$ (s):	*1.378*	*1.011*
Mean wave period $T_{0,2}$ (s):	*1.286*	*0.8541*
Spectral width (Broadness):	*0.7892*	*0.8278*
Spectral width (Narrowness):	*0.3858*	*0.6344*

Time Domain Analysis

Overall amplitude reflection coefficient: *34.2%*

	Incident	Reflected
Number of waves:	942	1230
Mean wave height H_m (m):	*0.1049*	*0.03537*
Mean wave period T_m (s):	*1.452*	*1.112*
Sig. wave height H_s (m):	*0.1519*	*0.05348*
$T_{H_{1/3}}$ (s):	*1.585*	*1.423*
H_{max} (m):	*0.2284*	*0.08686*
T_{Hmax} (s):	*1.645*	*1.788*
$H_{1/10}$ (m):	*0.1764*	*0.06524*
$H_{1/50}$ (m):	*0.202*	*0.07832*
$H_{1/100}$ (m)	*0.2134*	*0.08228*
$H_{1/250}$ (m):	*0.2227*	*0.08536*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08684*
$H_{1\%}$ (m):	*0.1994*	*0.07819*
$H_{2\%}$ (m):	*0.1859*	*0.07167*
$H_{10\%}$ (m)	*0.1595*	*0.05632*
Groupiness factor GF:	*0.848*	*1.004*
Skewness b1:	*0.5568*	*0.2073*
Kurtosis b2:	*2.66*	*3.021*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-11 Emerged Berm 4% - Test 2.experimentvedpoint.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.469 Hz*
Freq. upper bound:	*9.375 Hz*

Freq. Domain Analysis

Overlap (FFT) timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*19*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*20*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *28.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.07232*	*0.02082*
Peak wave period T_P (s):	*1.067*	*1.123*
Energy wave period $T_{-1,0}$ (s):	*1.054*	*0.971*
Mean wave period $T_{0,1}$ (s):	*1.021*	*0.8502*
Mean wave period $T_{0,2}$ (s):	*0.9996*	*0.7784*
Spectral width (Broadness):	*0.5055*	*0.7411*
Spectral width (Narrowness):	*0.2063*	*0.4391*

Time Domain Analysis

Overall amplitude reflection coefficient: *28.9%*

	Incident	**Reflected**
Number of waves:	1006	1162
Mean wave height H_m (m):	*0.04493*	*0.01343*
Mean wave period T_m (s):	*1.001*	*0.8667*
Sig. wave height H_s (m):	*0.07164*	*0.0205*
$T_{H_{1/3}}$ (s):	*1.049*	*0.9959*
H_{max} (m):	*0.1317*	*0.03906*
T_{Hmax} (s):	*1.051*	*0.7931*
$H_{1/10}$ (m):	*0.08835*	*0.02449*
$H_{1/50}$ (m):	*0.104*	*0.02895*
$H_{1/100}$ (m)	*0.1102*	*0.03083*
$H_{1/250}$ (m):	*0.12*	*0.03328*
$H_{0.1\%}$ (m):	*0.1317*	*0.0382*
$H_{1\%}$ (m):	*0.1011*	*0.02832*
$H_{2\%}$ (m):	*0.09642*	*0.02599*
$H_{10\%}$ (m)	*0.07699*	*0.02195*
Groupiness factor GF:	*0.9605*	*0.9456*
Skewness b1:	*0.03339*	*0.08339*
Kurtosis b2:	*2.921*	*2.824*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-11 Emerged Berm 4% - Test 2.experimentveldata.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.406 Hz*
Freq. upper bound:	*8.125 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap (Fourier) timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*21*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*24*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *32%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1056*	*0.03381*
Peak wave period T_P (s):	*1.231*	*1.231*
Energy wave period $T_{-1,0}$ (s):	*1.211*	*1.064*
Mean wave period $T_{0,1}$ (s):	*1.156*	*0.8879*
Mean wave period $T_{0,2}$ (s):	*1.116*	*0.795*
Spectral width (Broadness):	*0.6627*	*0.7832*
Spectral width (Narrowness):	*0.2687*	*0.4972*

Time Domain Analysis

Overall amplitude reflection coefficient: *32.1%*

	Incident	Reflected
Number of waves:	992	1215
Mean wave height H_m (m):	*0.06601*	*0.02149*
Mean wave period T_m (s):	*1.137*	*0.9282*
Sig. wave height H_s (m):	*0.1028*	*0.03321*
$T_{H_{1/3}}$ (s):	*1.227*	*1.102*
H_{max} (m):	*0.1494*	*0.06032*
T_{Hmax} (s):	*1.193*	*0.6622*
$H_{1/10}$ (m):	*0.1224*	*0.04111*
$H_{1/50}$ (m):	*0.1372*	*0.04949*
$H_{1/100}$ (m)	*0.1416*	*0.05193*
$H_{1/250}$ (m):	*0.1456*	*0.05508*
$H_{0.1\%}$ (m):	*Not enough data*	*0.05929*
$H_{1\%}$ (m):	*0.136*	*0.04867*
$H_{2\%}$ (m):	*0.1313*	*0.04545*
$H_{10\%}$ (m)	*0.1099*	*0.03511*
Groupiness factor GF:	*0.9035*	*1.034*
Skewness b1:	*0.1557*	*0.1821*
Kurtosis b2:	*2.812*	*3.08*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-11 Emerged Berm 4% - Test 2.experimentvelocity.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *30.8%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1643*	*0.05054*
Peak wave period T_P (s):	*1.73*	*1.939*
Energy wave period $T_{-1,0}$ (s):	*1.556*	*1.351*
Mean wave period $T_{0,1}$ (s):	*1.404*	*0.924*
Mean wave period $T_{0,2}$ (s):	*1.297*	*0.7774*
Spectral width (Broadness):	*0.8058*	*0.8055*
Spectral width (Narrowness):	*0.4127*	*0.6423*

Time Domain Analysis

Overall amplitude reflection coefficient: *30.6%*

	Incident	Reflected
Number of waves:	952	1376
Mean wave height H_m (m):	*0.1119*	*0.03349*
Mean wave period T_m (s):	*1.499*	*1.037*
Sig. wave height H_s (m):	*0.158*	*0.05042*
$T_{H_{1/3}}$ (s):	*1.635*	*1.324*
H_{max} (m):	*0.2118*	*0.09279*
T_{Hmax} (s):	*1.541*	*1.809*
$H_{1/10}$ (m):	*0.1818*	*0.06132*
$H_{1/50}$ (m):	*0.2012*	*0.07146*
$H_{1/100}$ (m)	*0.2049*	*0.07654*
$H_{1/250}$ (m):	*0.2085*	*0.08323*
$H_{0.1\%}$ (m):	*Not enough data*	*0.09129*
$H_{1\%}$ (m):	*0.2*	*0.06942*
$H_{2\%}$ (m):	*0.1949*	*0.06505*
$H_{10\%}$ (m)	*0.1658*	*0.05403*
Groupiness factor GF:	*0.8116*	*1.013*
Skewness b1:	*0.5732*	*0.1335*
Kurtosis b2:	*2.576*	*3.104*